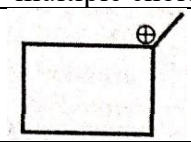
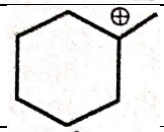
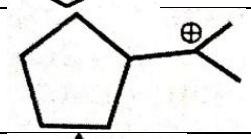
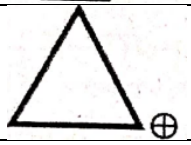
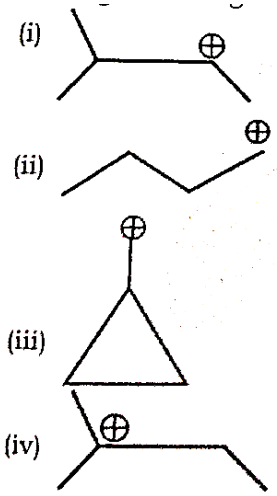
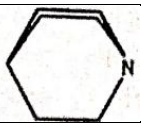
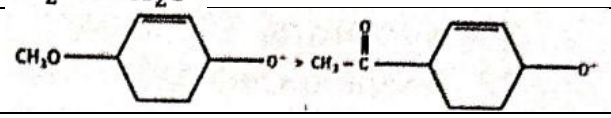
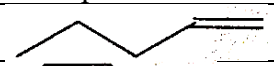
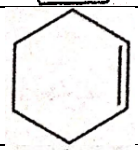
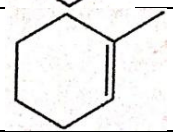
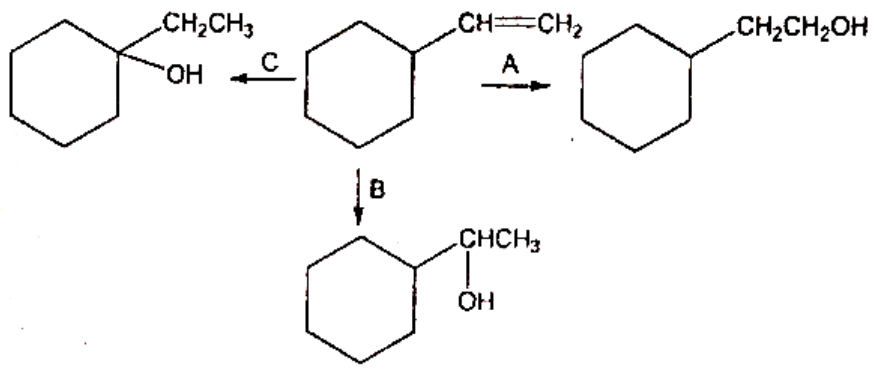
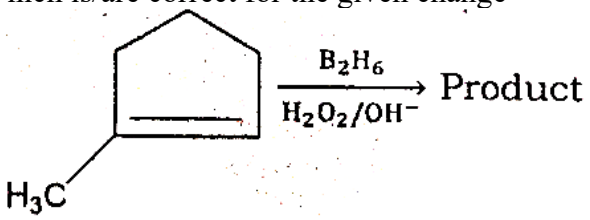
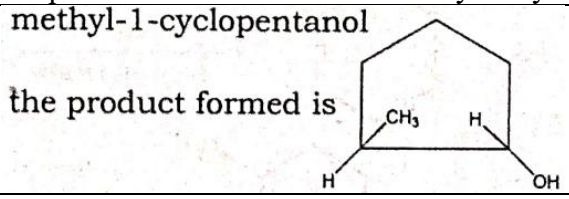
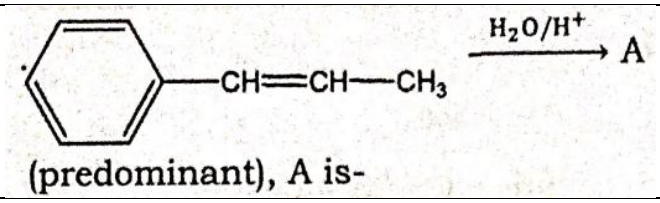
Question	Which behaves both as a nucleophile and electrophile?
Type	multiple choice
Option	CH_3NH_2
Option	CH_3Cl
Option	CH_3CN
Option	CH_3OH
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following structure can undergo ring expansion?
Type	multiple choice
Option	
Option	
Option	
Option	
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following carbocation can undergo rearrangement? 
Type	multiple choice
Option	(i) only
Option	(i), (ii) and (iii)
Option	(iii) and (iv) only
Option	(i) and (ii) only
Answer	4
Solution	
Positive Marks	4
Negative Marks	1

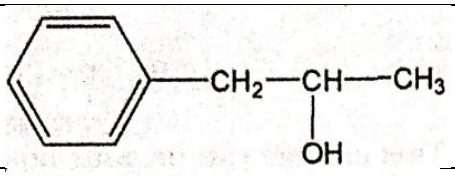
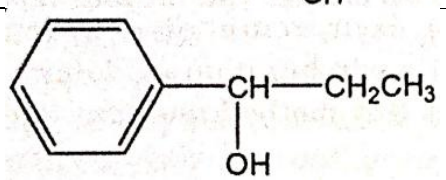
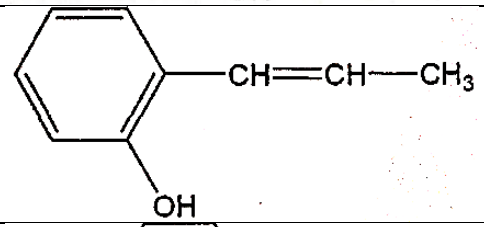
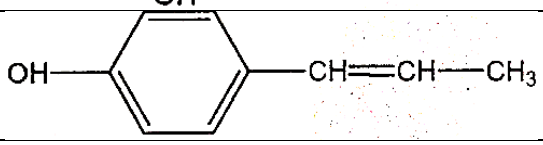
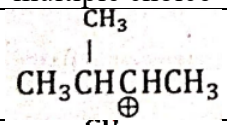
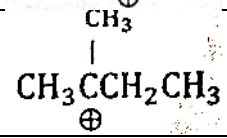
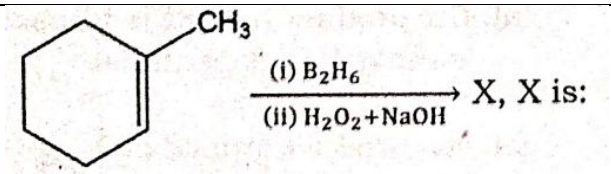
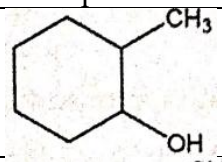
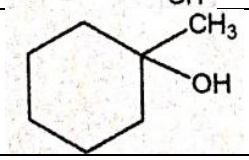
Question	Which of the following is correct order of nucleophilicity?
Type	multiple choice
Option	$(CH_3CH_2)_3N >$ 
Option	$HO^- > H_2N^-$
Option	$H_2O > H_2S$
Option	
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	In the following cation----------migrates to a positive charged carbon atom.
Type	multiple choice
Option	$CH_3\ at\ C-4$
Option	$H\ at\ C-2$
Option	$CH_3\ at\ C-2$
Option	$H\ \ at\ C-4$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is strongest nucleophile?
Type	multiple choice
Option	COO^{--}
Option	$\ddot{O}H$
Option	$\ddot{C}H$
Option	$C_2H_5\ddot{O}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following groups contains only electrophiles?
Type	multiple choice
Option	BF_3, NH_3, H_2O
Option	$AlCl_3, SO_4\overset{\oplus}{N}O_2$
Option	$\overset{\oplus}{N}O_2, \overset{\oplus}{C}H_3\overset{\oplus}{N}a$
Option	$\bar{C}_2H_5, \overset{\odot}{C}_2H_5, \overset{\oplus}{C}_2H_5$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following are nucleophile but not electrophile?
Type	multiple choice
Option	H_2O

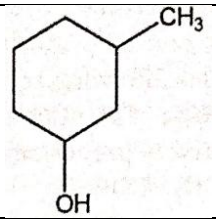
Option	$H_3O^{\oplus}$
Option	$\overset{\oplus}{N}H_4$
Option	BH_3
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The increasing order of nucleophilicity of the following nucleophiles is: (1) $CH_3CO_2^{\ominus}$ (2) H_2O (3) $CH_3SO_3^{\ominus}$ (4) $\overset{\ominus}{O}H$
Type	multiple choice
Option	2 < 3 < 4 < 1
Option	2 < 3 < 1 < 4
Option	1 < 4 < 3 < 2
Option	4 < 1 < 3 < 2
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is an electrophile?
Type	multiple choice
Option	H_2O
Option	SO_3
Option	NH_3
Option	ROR
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which one gives only one monosubstitution product on chlorination -
Type	multiple choice
Option	n-Pentane
Option	Neopentane
Option	Isopentane
Option	n-Butane
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	When an aqueous solution containing sodium acetate and sodium acetate and sodium propionate is electrolysed we get
Type	multiple choice
Option	Ethane
Option	Propane
Option	Butane
Option	All
Answer	4
Solution	

Positive Marks	4
Negative Marks	1
Question	$C_5H_{11}Cl$ by Wurtz reaction forms 2,2,5,5-tetramethyl hexane as the main product. Hence $C_5H_{11}Cl$ is -
Type	multiple choice
Option	1-Chloro -2-2-dimethylpropane
Option	2-Methyl-1-chlorobutane
Option	Both are true
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Aq. Solution of potassium acetate is electrolysed. Possible organic product(s) are-
Type	multiple choice
Option	CH_3CH_3
Option	CH_3COOCH_3
Option	$CH_3CH_2CH_2CH_3$
Option	(a) and (b)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following give propane on decarboxylation-
Type	multiple choice
Option	n-Butyric acid
Option	Isobutyric acid
Option	Both (a) & (b)
Option	None of these
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following compound. is produced when $CH_2 = CH - (CH_2)_5 COOH$ reacts with HBr is presence of peroxides
Type	multiple choice
Option	$CH_3CH(CH_2)_5 COOH$
Option	$BrCH_2CH_2(CH_2)_5 COOH$
Option	$CH_3CH_2CH_2(CH_2)_5 COOH$
Option	$CH_3CH_2Br(CH_2)_5 COOH$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Addition of HCl to vinyl chloride gives 1,1-dichloroethane because of-
Type	multiple choice
Option	Mesomeric effect of C1
Option	Inductive effect of Cl

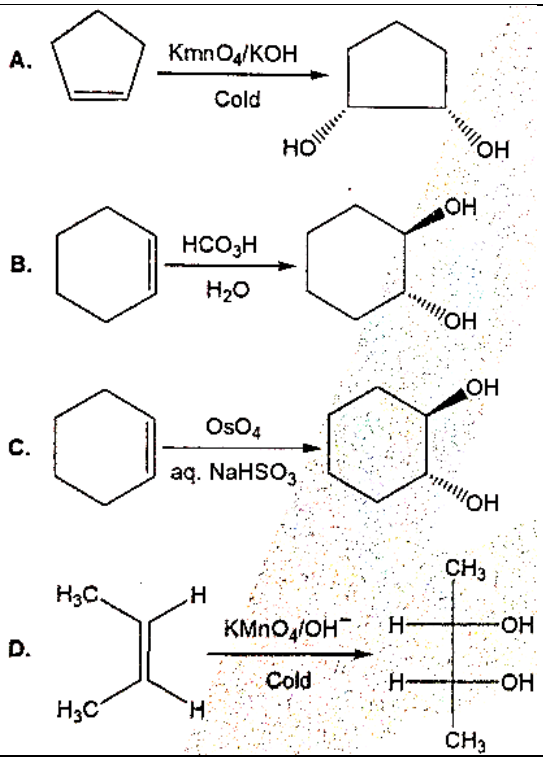
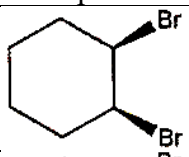
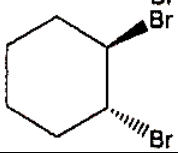
Option	Restricted rotation around double bond
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which doesn't follow Markovnikov's rule
Type	multiple choice
Option	$CH_3-CH=CH_2$
Option	$CH_3CH=CHCH_3$
Option	$CH_3-\underset{\underset{CH_3}{\vert}}{CH}-CH=CH_2$
Option	$CH_3-CH_2-CH=CH_2$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	HBO, oxymercuration demercuration and acid catalyzed hydration will give same product in-
Type	multiple choice
Option	
Option	
Option	
Option	
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	$CH_3-\underset{\underset{CH_3}{\vert}}{\overset{\overset{CH_3}{\vert}}{C}}-CH=CH_2 \longrightarrow CH_3-\underset{\underset{CH_3}{\vert}}{\overset{\overset{CH_3}{\vert}}{C}}-\underset{\underset{OH}{\vert}}{CH}-CH_3$ This change can be done by-
Type	multiple choice
Option	acid catalysed hydration
Option	oxymercuration-demercuration
Option	hydroboration-oxidation
Option	any method mentioned above
Answer	2
Solution	

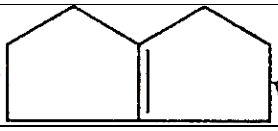
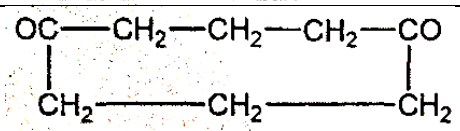
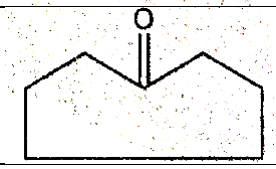
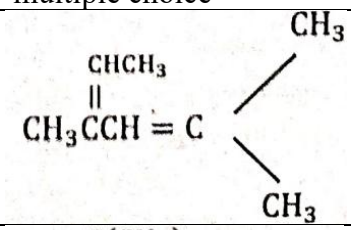
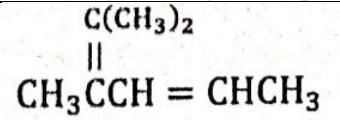
Positive Marks	4
Negative Marks	1
Question	 Schemes A, B, C are I. acid catalysed hydration II. HBO III. oxymercuration-demercuration
Type	multiple choice
Option	I in all cases
Option	I, II, III
Option	II, III, I
Option	III, I, III
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which is/are correct for the given change 
Type	multiple choice
Option	the product formed is trans-2- methyl-1-cyclopentanol
Option	
Option	the addition is syn addition
Option	all of the above
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	
Type	multiple choice

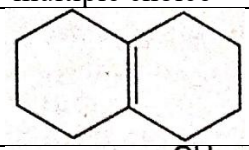
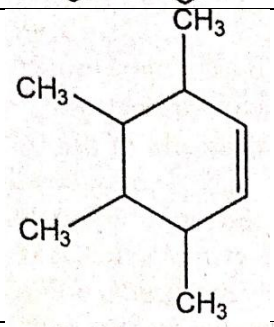
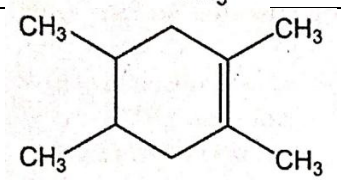
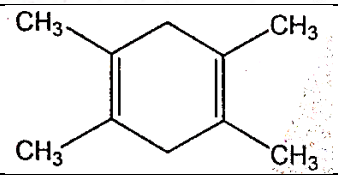
Option	
Option	
Option	
Option	
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Most stable intermediate of the hydration of isopentane $\begin{array}{l} CH_3CHCH=CH_2 \\ \quad\;\; \vert \\ \quad\;\; CH_3 \end{array}$ is
Type	multiple choice
Option	
Option	
Option	both are correct
Option	none is correct
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	
Type	multiple choice
Option	
Option	

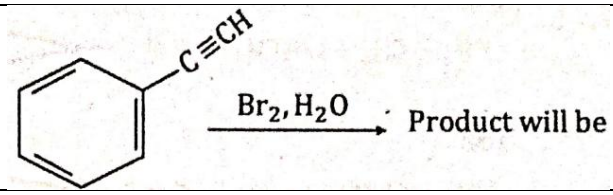
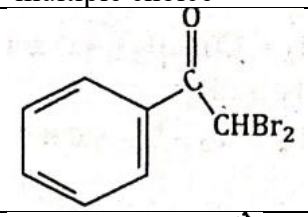
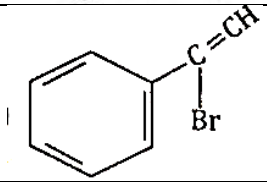
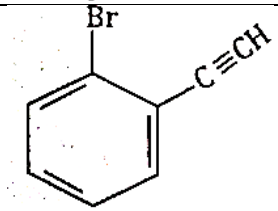
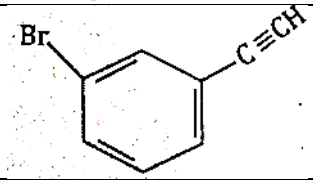
Option	
Option	None
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which can be used in Kolbe electrolysis-
Type	multiple choice
Option	CH_3CH_2COONa
Option	$\begin{array}{l} CH_2COONa \\ \vert \\ CH_2COONa \end{array}$
Option	$\begin{array}{l} CHCOONa \\ \Vert \\ CHCOONa \end{array}$
Option	All of these
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following statement is correct?
Type	multiple choice
Option	Fluorination is too violent to be controlled.
Option	Iodination is very slow and a reversible reaction.
Option	Iodination carried out in the presence of oxidizing agents (HIO_4 / HNO_3).
Option	All statements are correct.
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following oxidation reaction is/are correct?
Type	multiple choice
Option	$2CH_4 + O_2 \xrightarrow{Cu/523K/100atm} 2CH_3 - OH$
Option	$CH_4 + O_2 \xrightarrow{Mo_2O_3, \Delta} HCHO + H_2O$
Option	$2CH_3 - CH_3 + 3O_2 \xrightarrow{(CH_3COO)_2Mn, \Delta} 2CH_3COOH + 2H_2O$
Option	All are correct
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	$\underset{\text{n-hexane}}{CH_3(CH_2)4CH_3} \rightarrow \underset{\text{2-methyl pentane}}{CH_3 - \underset{\vert \atop CH_3}{CH} - (CH_2)_2 - CH_3} + \underset{\text{3-methyl pentane}}{CH_3CH_2 - \underset{\vert \atop CH_3}{CH} - CH_2 - CH_3}$
Type	multiple choice

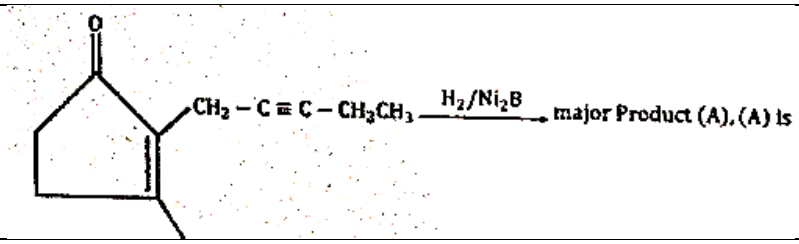
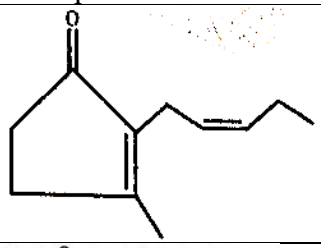
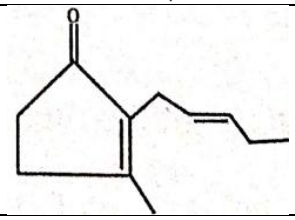
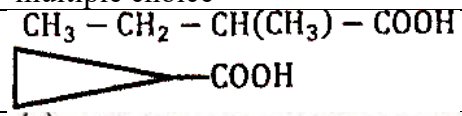
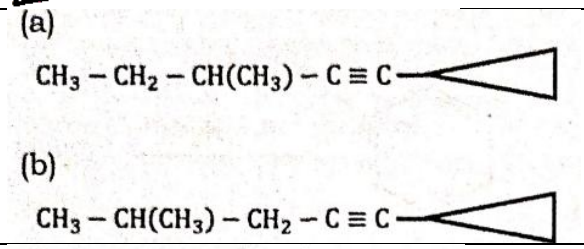
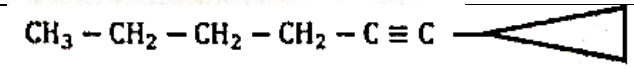
Option	Aromatization
Option	Pyrolysis
Option	Isomerization
Option	Oxidation
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Following reaction is known as $C_{10}H_{22} \xrightarrow{900\ K} C_4H_8 + C_6H_{14}$
Type	multiple choice
Option	Alkylation
Option	Cracking
Option	Dehydrogenation
Option	HydrolysisType here
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Oxymercuration-demercuration reaction of 1-methylcyclohexene gives mainly
Type	multiple choice
Option	Cis-2-Methylcyclohexanol
Option	Trans-2-Methylcyclohexanol
Option	1-Methylcyclohexanol
Option	Mixture of cis and
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Propylene on hydrolysis with dilute sulphuric acid forms:
Type	multiple choice
Option	n-Propyl alcohol
Option	Isopropyl alcohol
Option	Ethyl alcohol
Option	Butyl alcohol
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which is/are incorrect reaction(s)-

	
Type	multiple choice
Option	Only A and B
Option	Only A and C
Option	Only C
Option	Only B and C
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	+ $Br_2 \xrightarrow{CCl_4}$ Major product (A), (A) will have configuration-
Type	multiple choice
Option	
Option	
Option	Both (a) and (b)
Option	None of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	$CCl_3CH=CH_2 \xrightarrow{Cl_2+H_2O} A,\ A$ is
Type	multiple choice
Option	$CCl_3CH(OH)CH_2Cl$

Option	$CCl_3CH(Cl)CH_2OH$
Option	$CCl_3CH(Cl)CH_2Cl$
Option	$CCl_3CH(OH)CH_2(OH)$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which is most suitable reaction for the synthesis of only 2-bromobutane
Type	multiple choice
Option	$CH_3CH{=}CHCH_3 \xrightarrow{HBr}$
Option	$CH_3CH_2CH{=}CH_2 \xrightarrow{HBr}$
Option	$CH_3CH{=}CHCH_3 \xrightarrow{HBr}$
Option	$CH_3CH_2CH{=}CH_2 \xrightarrow[HBr]{Peroxide}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Ozonolysis of  will give-
Type	multiple choice
Option	
Option	
Option	
Option	None of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Alkene $\xrightarrow{O_3/H_2O}$ CH_3COCH_3 + CH_3COOH + $CH_3COCOOH$
Type	multiple choice
Option	
Option	

Option	Both are correct
Option	None is correct
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	$C_{10}H_{16}(A)$ changes $C_{10}H_{16}O_2$ (B) only ozonolysis. If B is symmetrical then (A)
Type	multiple choice
Option	
Option	
Option	
Option	
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	$(CH_3)_2C{=}CH_2 \xrightarrow{KMnO_4/H^+} A + B$
Type	multiple choice
Option	$CH_3{-}\underset{\underset{O}{\|}}{C}{-}CH_3$, HCOOH
Option	$CH_3{-}\underset{\underset{O}{\|}}{C}{-}CH_3$, CO_2
Option	$CH_3{-}CH_2{-}COOH$, H–COOH
Option	$CH_3{-}COOH$, HCOOH
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Identify the incorrect statements (i) Alkynes are more reactive than alkenes towards electrophilic addition reaction (ii) Alkynes are less reactive than alkenes towards electrophilic addition reaction (iii) Alkynes decolourise Br_2/CCl_4

	(iv) Addition of HBr to alkynes in presence of peroxide proceeds via Markovnikov's rule
Type	multiple choice
Option	(i) and (ii)
Option	(ii) and (iii)
Option	(i) and (iv)
Option	(ii) and (iv)
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which alkyne produces pyruvic acid with cold dil. Alkaline $KMnO_4$?
Type	multiple choice
Option	$CH \equiv CH$
Option	$CH_3C \equiv CH$
Option	$CH_3C \equiv C - CH_3$
Option	$CH_3 - CH_2 - C \equiv CH$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	
Type	multiple choice
Option	
Option	
Option	
Option	
Answer	1
Solution	
Positive Marks	4
Negative Marks	1

Question	
Type	multiple choice
Option	
Option	
Option	Both (a) and (b)
Option	None of the above
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following alkynes will give following two products (A) and (B) on oxidative cleavage with hot $KMnO_4 / OH^-$:
Type	multiple choice
Option	
Option	
Option	$CH_3 - CH_2 - CH(CH_3) - C \equiv C - CH_2 - CH_2 - CH_3$
Option	
Answer	1
Solution	
Positive Marks	4
Negative Marks	1